Bachelor in Biomedical Engineering
Introduction to Bioengineering 25/26

Optical Imaging

Jesús Sánchez Cornago
jesussco@ing.uc3m.es

Daniel Sanderson Moreno
dsanders@ing.uc3m.es

Table of Contents

1. What is light?
    a. Light as a wave
    b. Light-matter interaction
2. Measuring spatial resolution
    a. Point Spread Function (PSF)
    b. Full Width Half Maximum (FWHM)
    c. Signal Saturation

What is light?

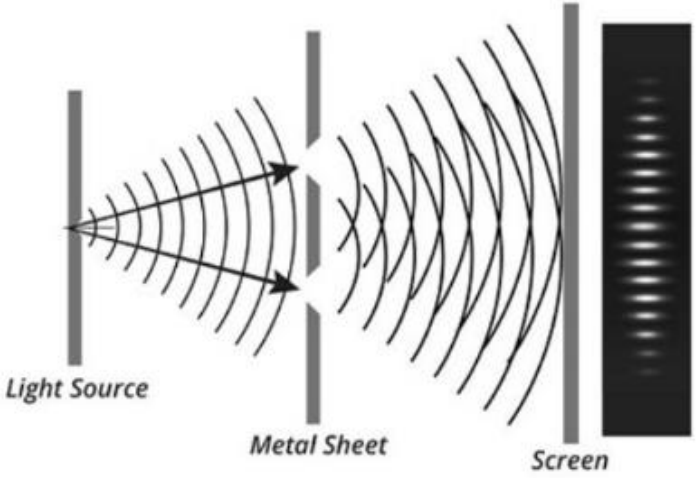
https://www.youtube.com/userQuantumFracture/

https://www.youtube.com/c/DateunVoltio

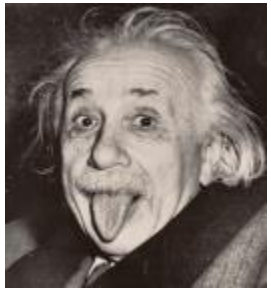


Thomas Young

Double Slit Experiment (1801)

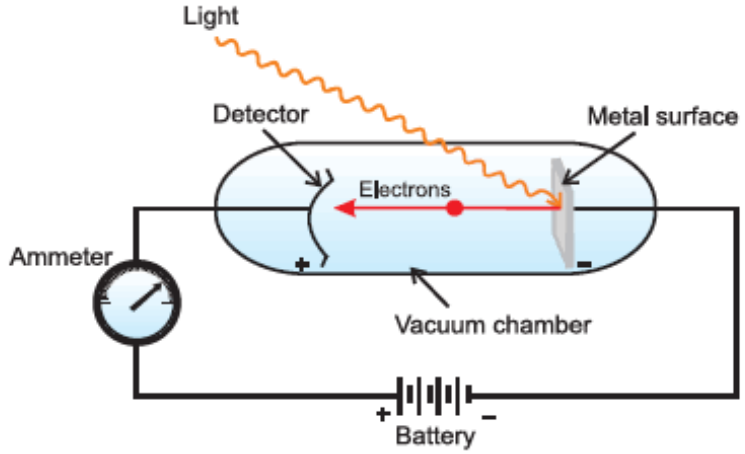


→ It's a wave



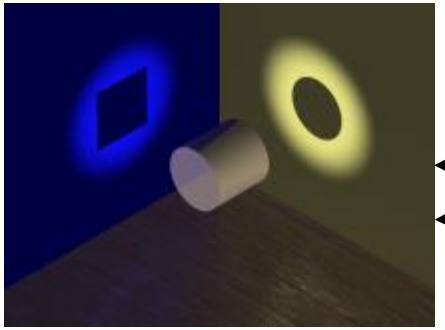
Albert Einstein

Photoelectric Effect Experiment (1905)



→ It's a particle: **photon**

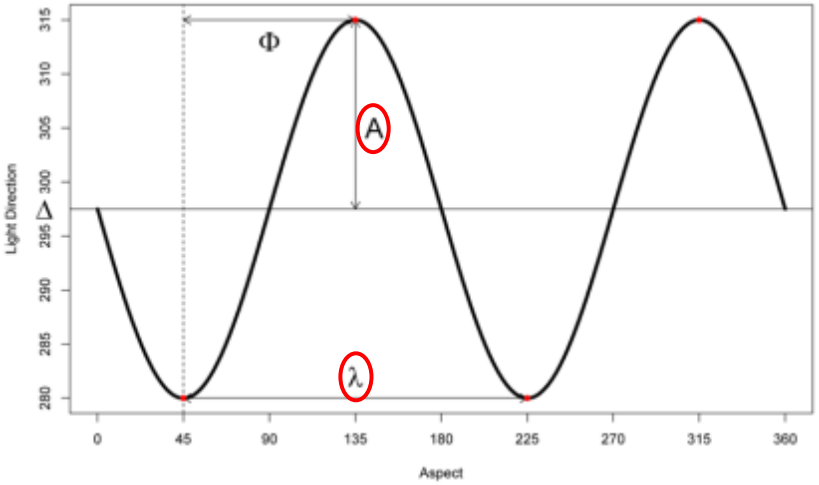
Wave-Particle Duality



Light as a wave

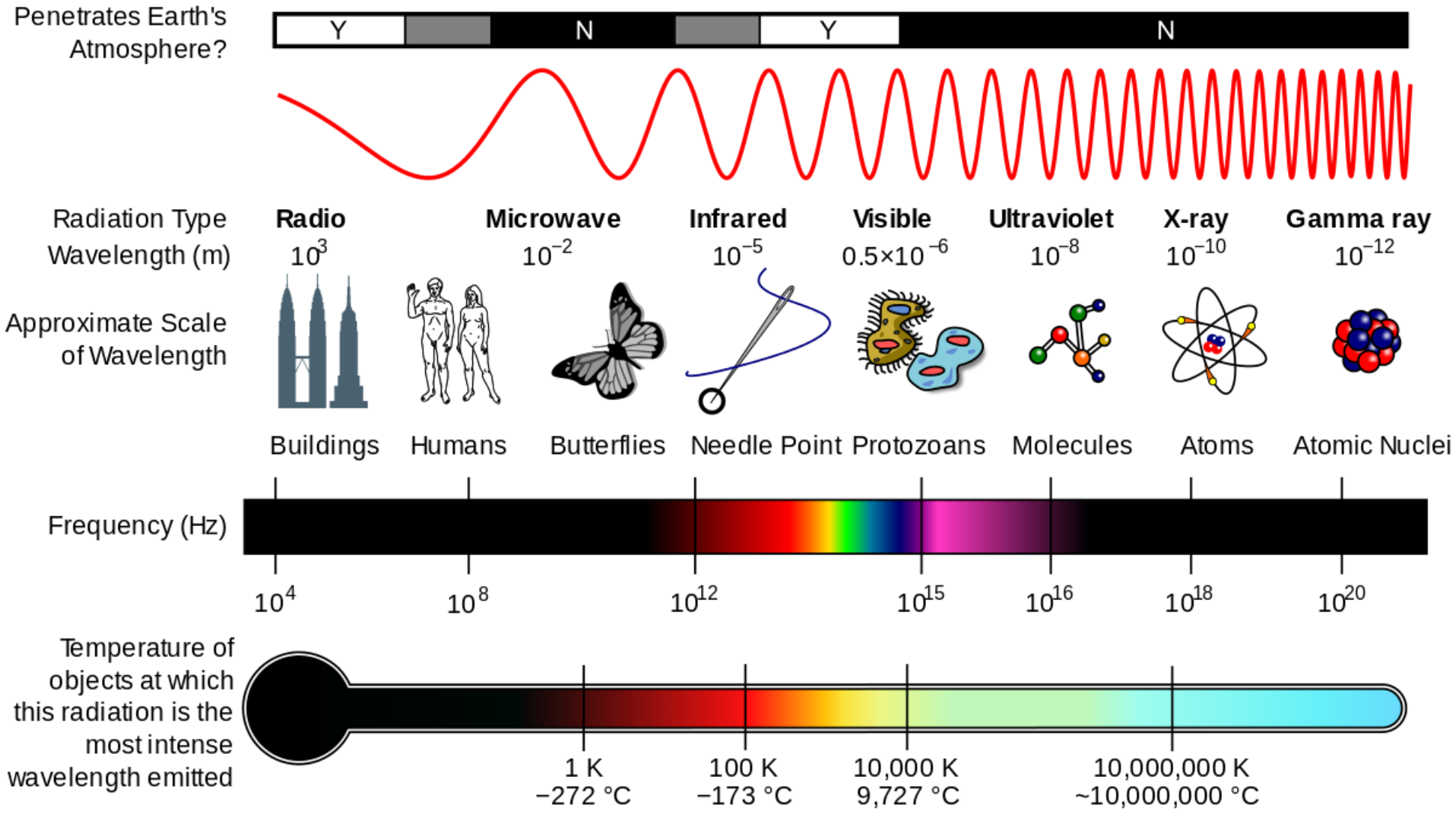


- Behavior akin to **water wavefronts**



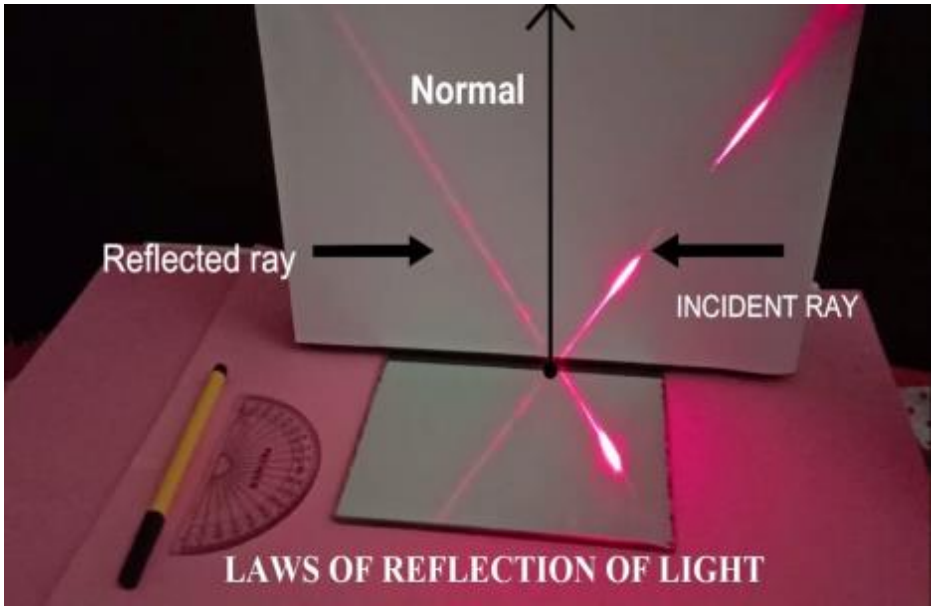
- **Amplitude (A):** Intensity
- **Wavelength (λ):** related to frequency

Light as a wave

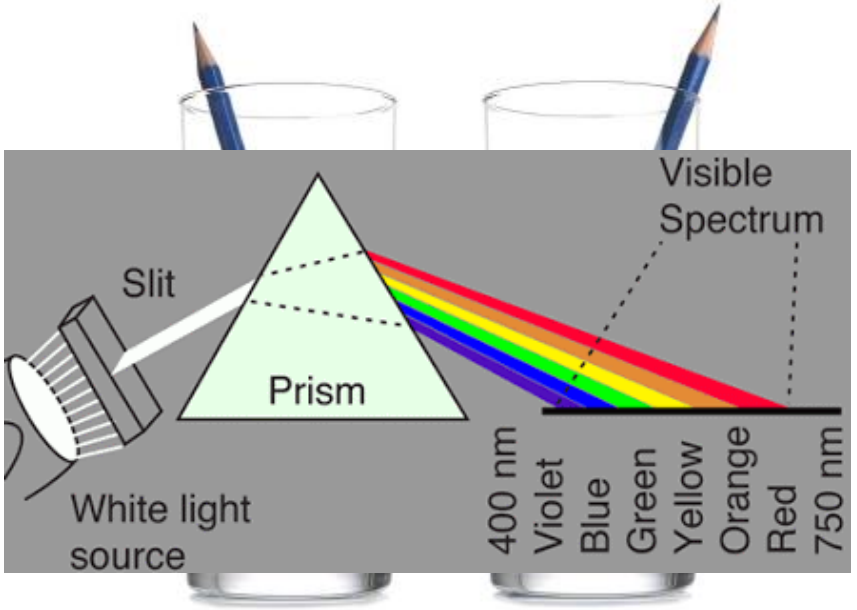


Light-matter interaction

Reflection: light bounces



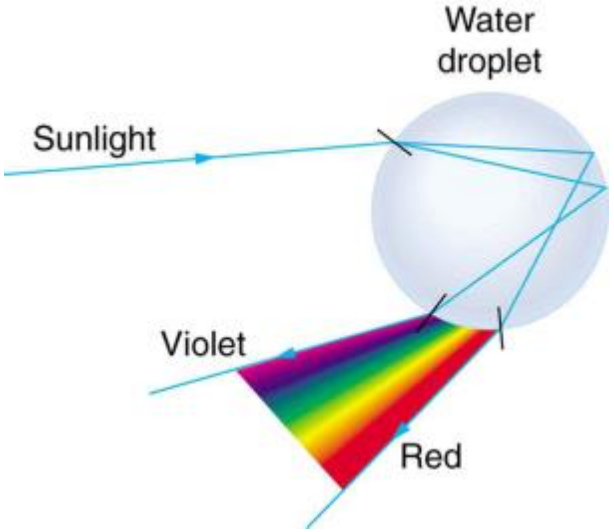
Refraction: light bends



Wavelength dependent!

Rainbow: several reflections and refractions in water droplets

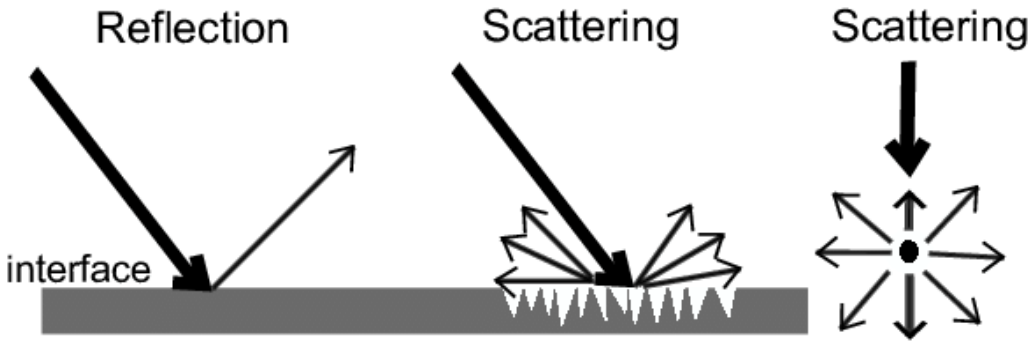




Scattering: “imperfect reflection”



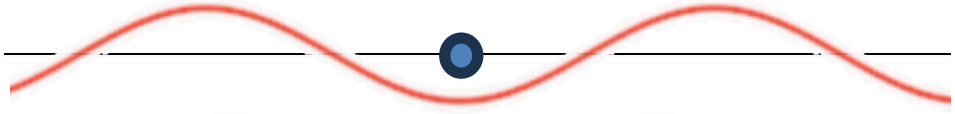
- Object geometry dependent

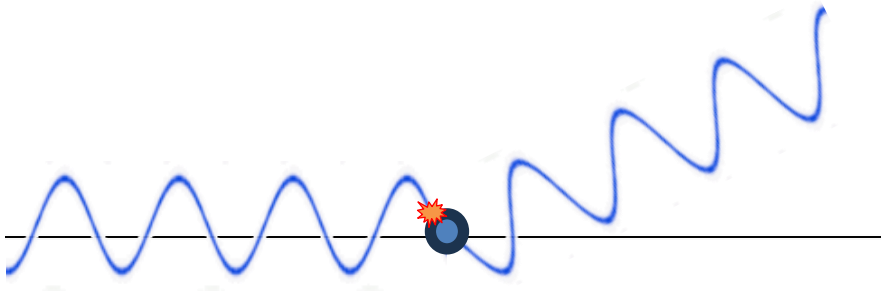


- Distance dependent

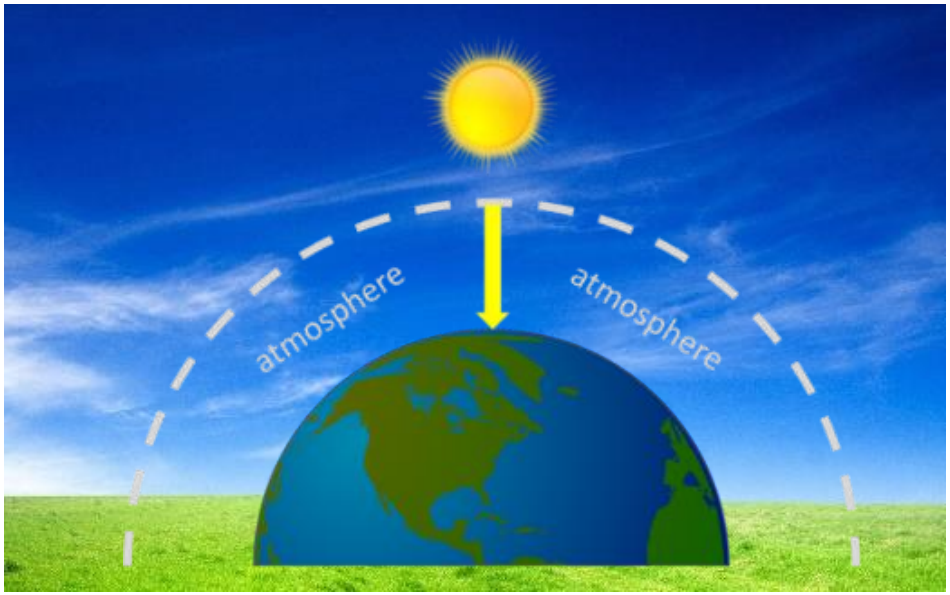


- Wavelength dependent

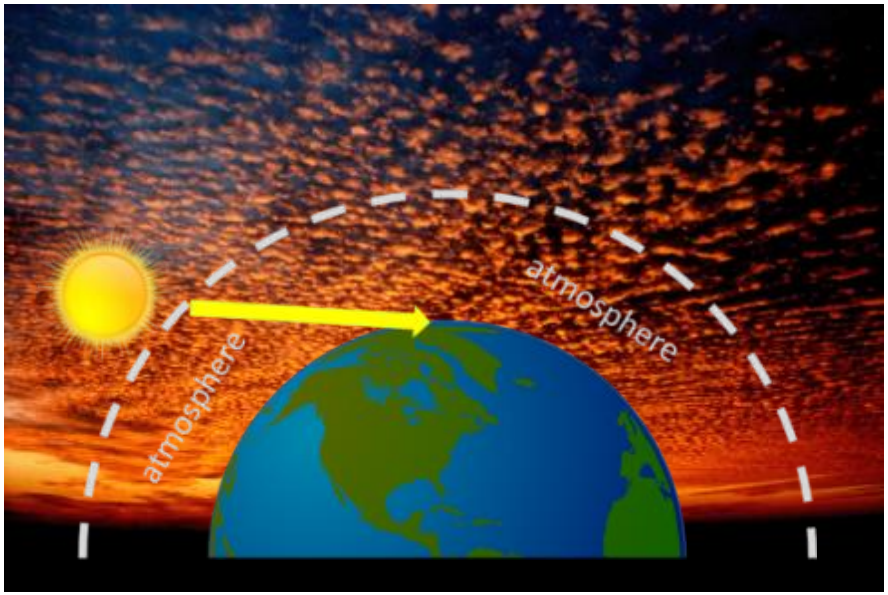




Day light



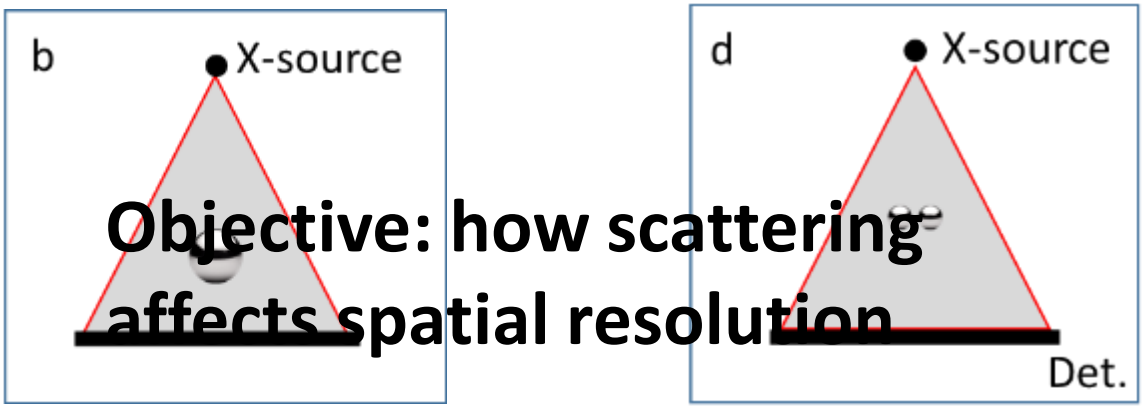
Sunset



Short wavelenghts (blue) are more scattered

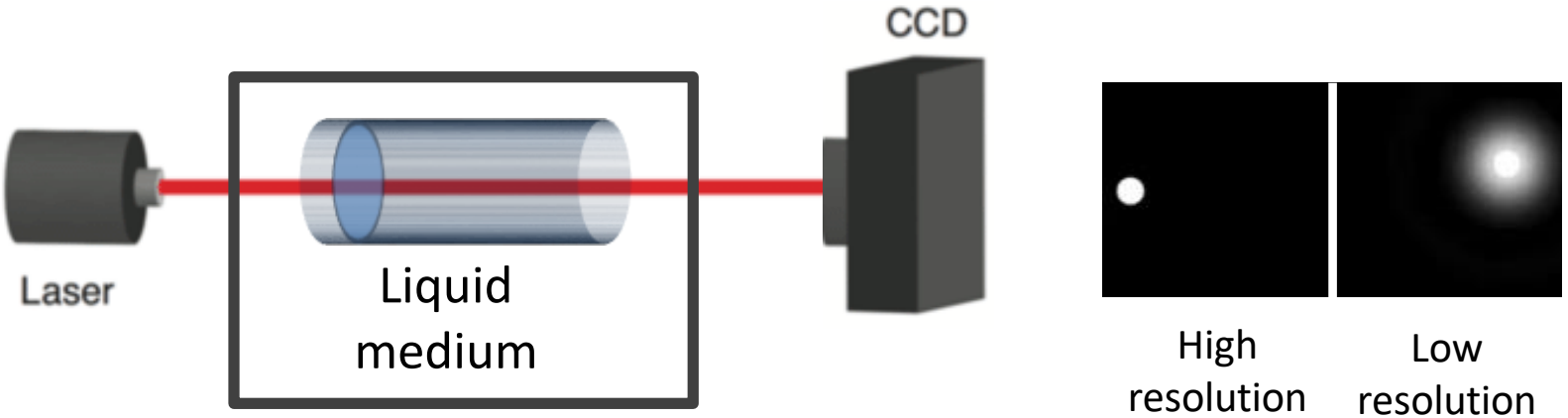
Resolution: Distance between two independent measures

One or two objects?



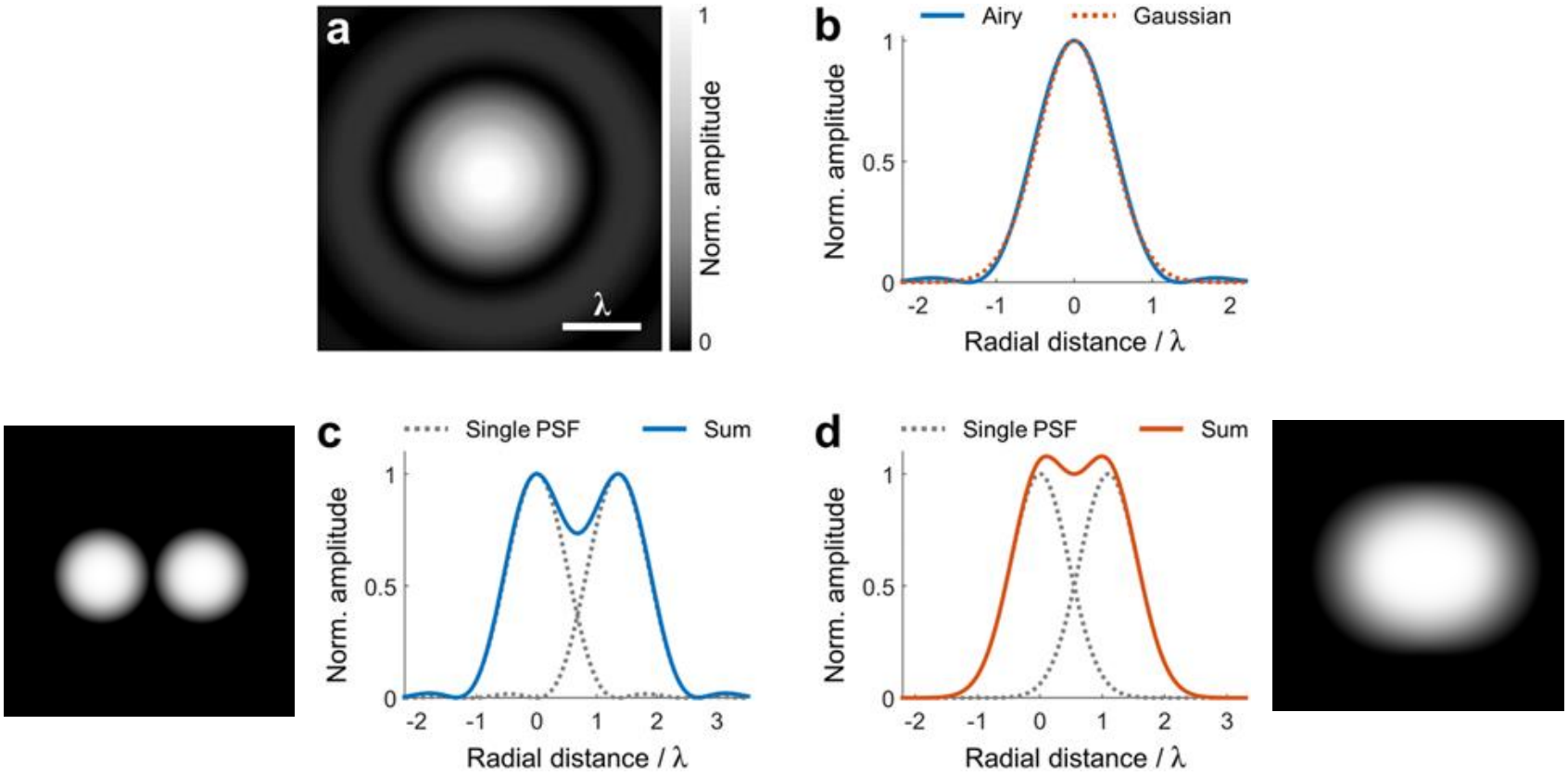
Objective: how scattering affects spatial resolution

Today: optical system's ability to detect a point source

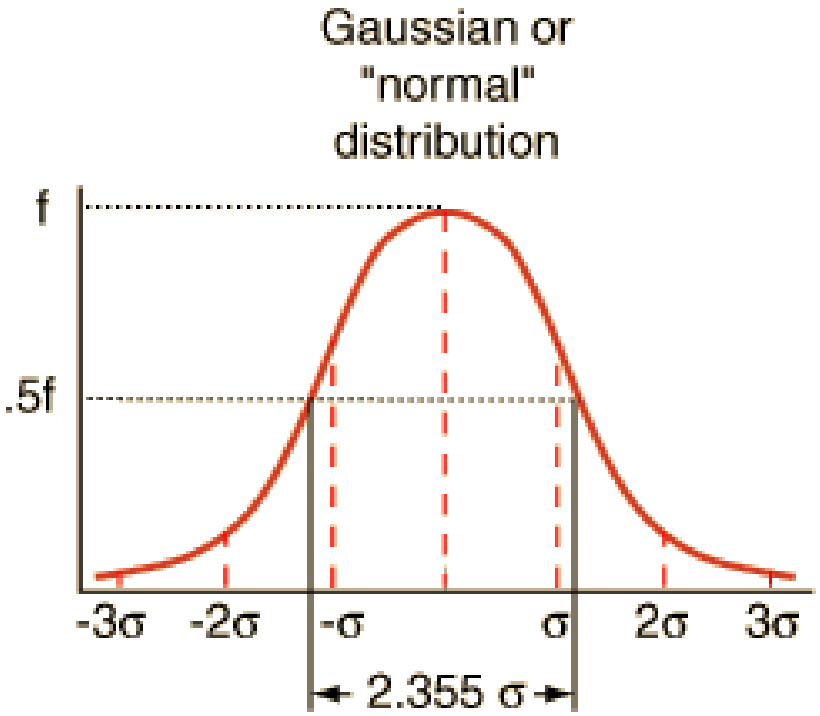


Point Spread Function (PSF)

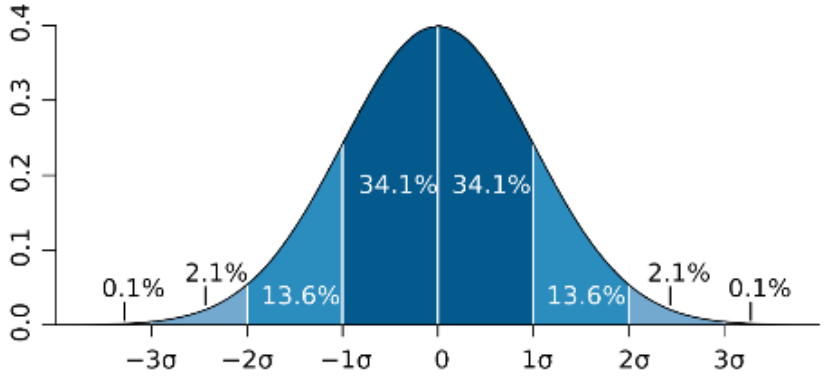
describes the response of a focused optical imaging system to a point source or point object.

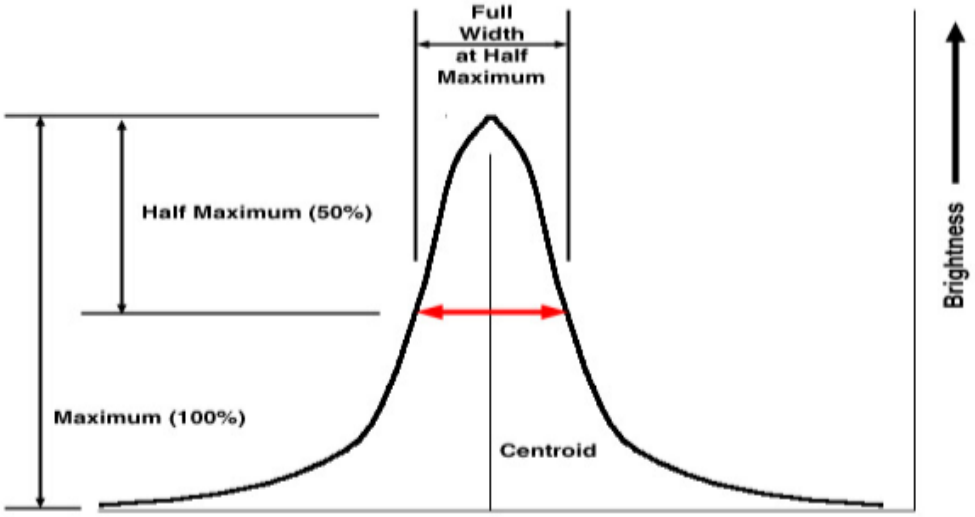


Full Width Half Maximum (FWHM)

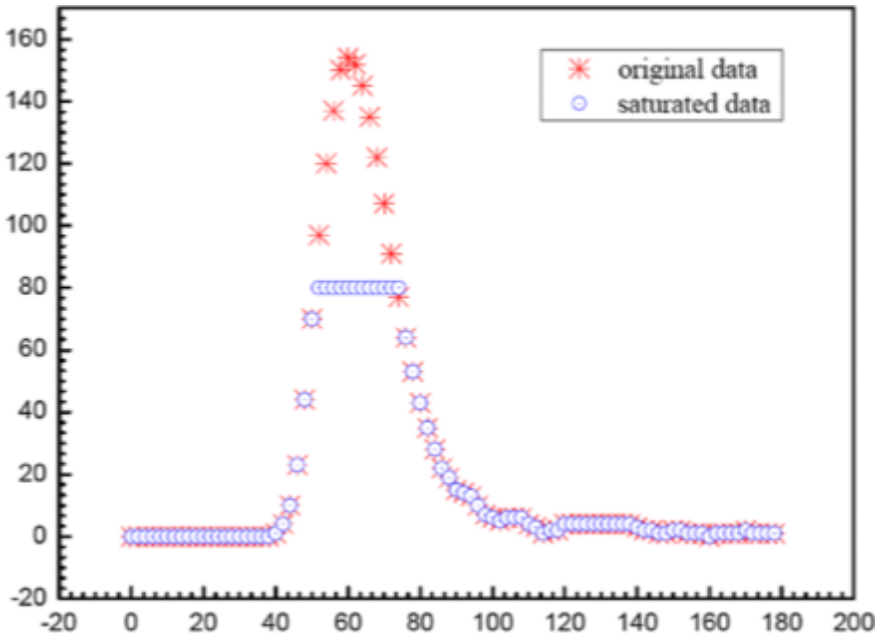


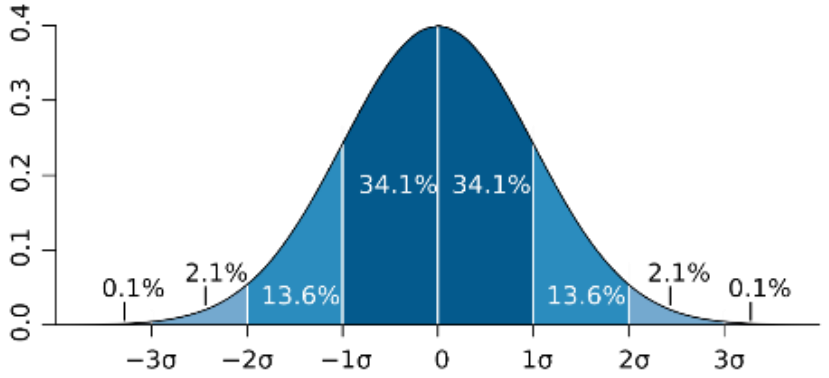
$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma \approx 2.355\,\sigma.$$



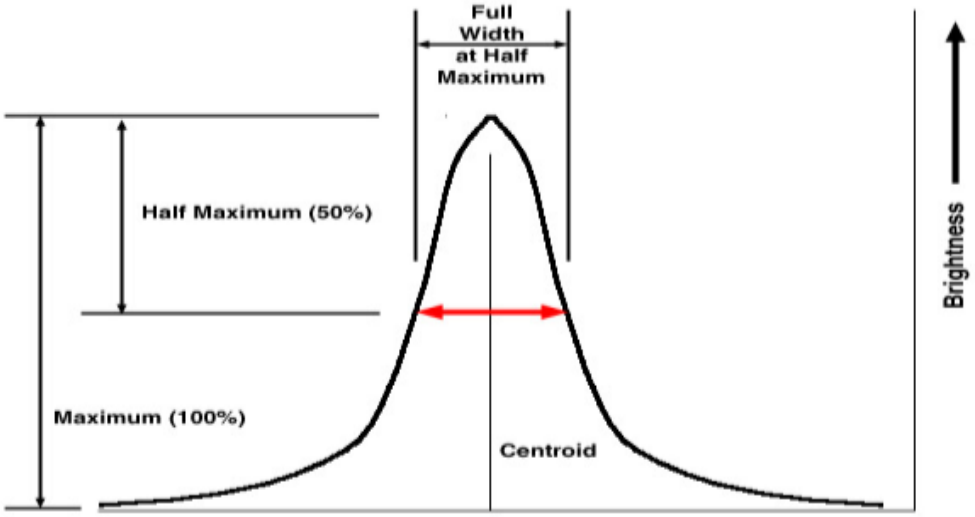


Signal saturation



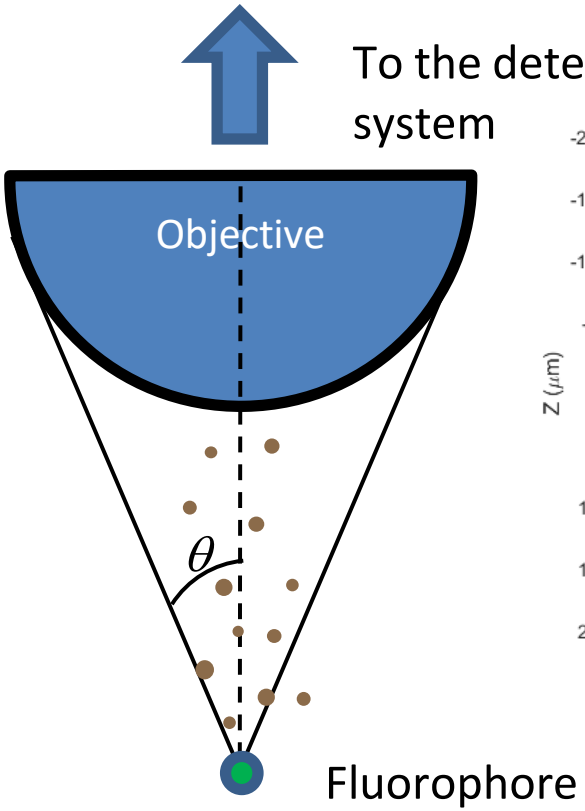


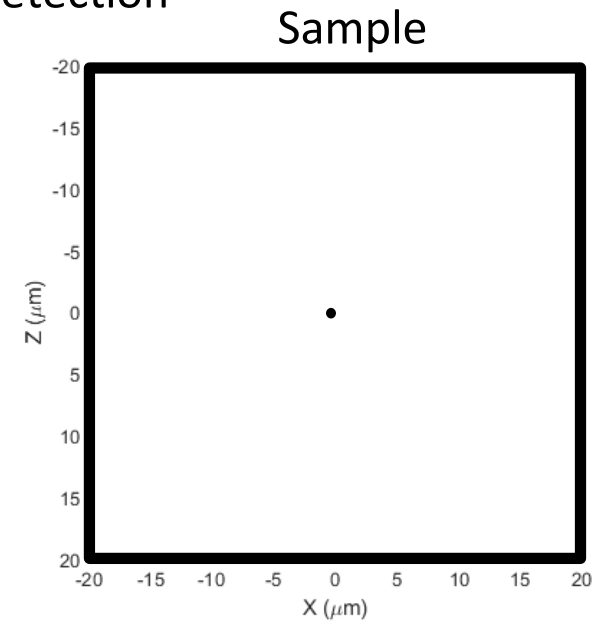
$$\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma \approx 2.355\,\sigma.$$

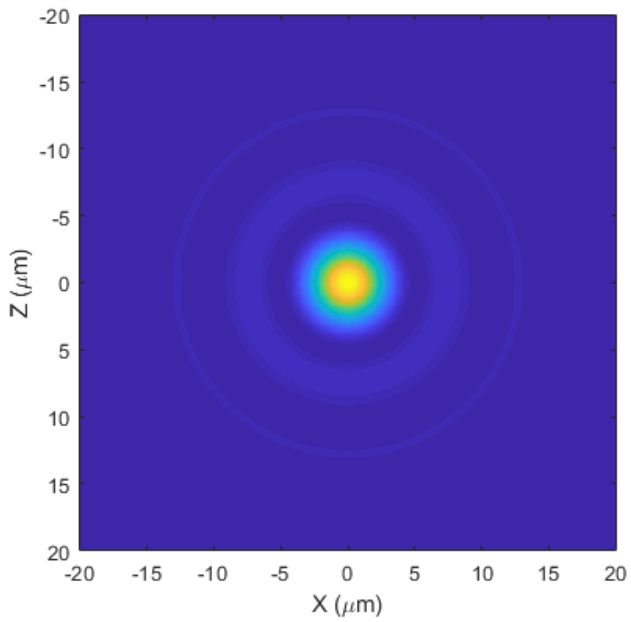


Kind reminder: real case example

Smallest fluorophore at the focal point in clear water gives this*; dirt particles (dots) would scatter its light:







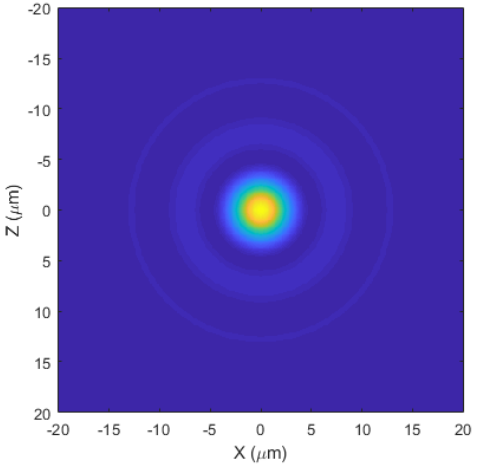
*in this lens system, of course

Dirt = suspended particles (polluted medium) that scatter light. In our practice: milk fat globules.

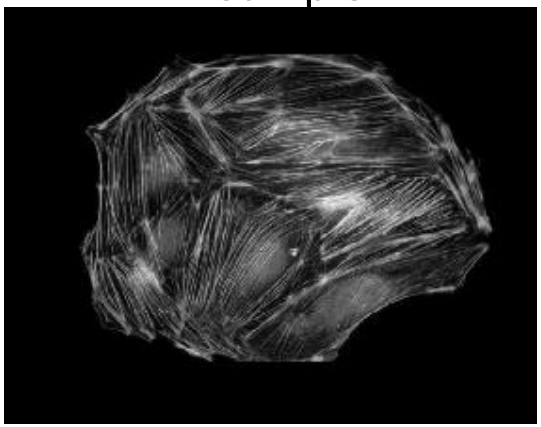
Kind reminder: real case example

In clear water the PSF stays narrow; dirt in suspension (our milk) widens it and the image blurs:

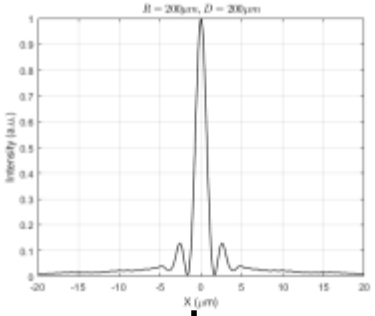
PSF



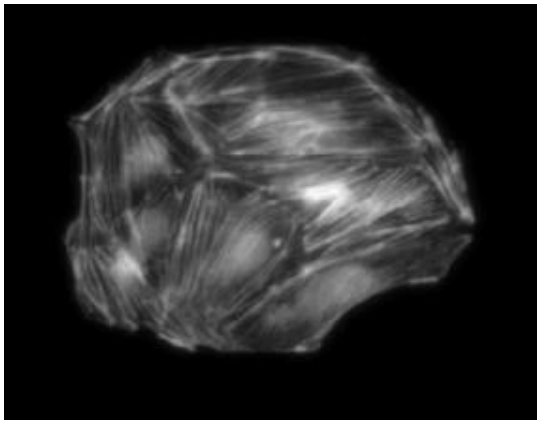
Sample



PSF narrow



Resulting image



PSF wide

